

Task 1:

```
package com.training.ooc;
interface LibraryUser {
void registerAccount();
void requestBook();
}
class KidUser implements LibraryUser {
int age;
String bookType;
KidUser(int age, String bookType) {
    this.age = age;
    this.bookType = bookType;
}
public void registerAccount() {
    if (age < 12) {
        System.out.println("You have successfully registered under a Kids Account");
    } else {
        System.out.println("Sorry, Age must be less than 12 to register as a kid");
    }
}
public void requestBook() {
    if (bookType.equalsIgnoreCase("Kids")) {
        System.out.println("Book Issued successfully, please return the book within 10 days");
    } else {
        System.out.println("You are allowed to take only kids books");
    }
}
}
class AdultUser implements LibraryUser {
int age;
String bookType;
AdultUser(int age, String bookType) {
    this.age = age;
    this.bookType = bookType;
}
public void registerAccount() {
    if (age > 12) {
        System.out.println("You have successfully registered under an Adult Account");
    } else {
        System.out.println("Sorry, Age must be greater than 12 to register as an adult");
    }
}
public void requestBook() {
    if (bookType.equalsIgnoreCase("Fiction")) {
        System.out.println("Book Issued successfully, please return the book within 7 days");
    } else {
        System.out.println("You are allowed to take only adult Fiction books");
    }
}
}
```

```
}  
public class program5 {  
public static void main(String[] args) {  
    LibraryUser kid = new KidUser(10, "Kids");  
    kid.registerAccount();  
    kid.requestBook();  
    LibraryUser adult = new AdultUser(23, "Fiction");  
    adult.registerAccount();  
    adult.requestBook();  
    LibraryUser wrongKid = new KidUser(14, "Fiction");  
    wrongKid.registerAccount();  
    wrongKid.requestBook();  
    LibraryUser wrongAdult = new AdultUser(9, "Kids");  
    wrongAdult.registerAccount();  
    wrongAdult.requestBook();  
}  
}
```

```
<terminated> program5 [Java Application] C:\Users\surya.p\p2\pool\plugins\org.eclipse  
You have successfully registered under a Kids Account  
Book Issued successfully, please return the book within 10 days  
You have successfully registered under an Adult Account  
Book Issued successfully, please return the book within 7 days  
Sorry, Age must be less than 12 to register as a kid  
You are allowed to take only kids books  
Sorry, Age must be greater than 12 to register as an adult  
You are allowed to take only adult Fiction books  
|
```

Task 2:

```
package com.training.ooc;  
import java.util.*;  
public class program6 {  
    public static ArrayList<Integer> fetchIndices(ArrayList<Integer> list1, ArrayList<Integer> list2) {  
        ArrayList<Integer> merged = new ArrayList<>(list1);  
        merged.addAll(list2);  
        Collections.sort(merged);  
        ArrayList<Integer> result = new ArrayList<>();  
        int[] indices = {2, 6, 8};  
        for (int idx : indices) {  
            if (idx < merged.size()) {  
                result.add(merged.get(idx));  
            }  
        }  
        return result;  
    }  
}
```

```
}  
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    ArrayList<Integer> list1 = new ArrayList<>();  
    ArrayList<Integer> list2 = new ArrayList<>();  
    for (int i = 0; i < 5; i++) {  
        list1.add(sc.nextInt());  
    }  
    for (int i = 0; i < 5; i++) {  
        list2.add(sc.nextInt());  
    }  
    ArrayList<Integer> finalList = fetchIndices(list1, list2);  
    System.out.println(finalList);  
}
```

```
<terminated> prog  
1  
2  
3  
4  
5  
11  
12  
13  
14  
15  
[3, 12, 14]
```

Task 3:

```
package com.training.ooc;  
import java.util.*;  
public class program7 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        HashMap<String, Float> studentMap = new HashMap<>();  
        System.out.println("Enter the number of students:");  
        int n = sc.nextInt();  
        sc.nextLine();  
        System.out.println("Enter student details (name and mark):");  
        for (int i = 0; i < n; i++) {  
            String name = sc.nextLine();  
            float mark = Float.parseFloat(sc.nextLine());  
            studentMap.put(name, mark);  
        }  
        System.out.println("Enter student name to get grade:");  
        String queryName = sc.nextLine();  
        if (studentMap.containsKey(queryName)) {
```

```
float mark = studentMap.get(queryName);  
if (mark < 60) {  
    System.out.println("Grade: FAIL");  
} else {  
    System.out.println("Grade: PASS");  
}  
} else {  
    System.out.println("Student not found.");  
}  
}  
}
```

```
Enter the number of students:  
2  
Enter student details (name and mark):  
surya  
95  
karthi  
90  
Enter student name to get grade:  
surya  
Grade: PASS  
|
```

Task 4:

```
package com.training.ooc;  
import java.util.*;  
public class program8 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        ArrayList<Integer> inputList = new ArrayList<>();  
        ArrayList<Integer> evenNumbersList = new ArrayList<>();  
        ArrayList<Integer> oddNumbersList = new ArrayList<>();  
        System.out.println("Enter integers (enter -1 to stop).");  
        while (true) {  
            int num = sc.nextInt();  
            if (num == -1) break;  
            inputList.add(num);  
        }  
        for (int num : inputList) {  
            if (num % 2 == 0) {  
                evenNumbersList.add(num);  
            } else {  
                oddNumbersList.add(num);  
            }  
        }  
        System.out.println("Input List: " + inputList);  
        System.out.println("Even Numbers List: " + evenNumbersList);  
    }  
}
```

```
System.out.println("Odd Numbers List: " + oddNumbersList);  
}  
}
```

```
Enter integers (enter -1 to stop):  
12  
1  
0  
8  
9  
11  
13  
67  
89  
-1  
Input List: [12, 1, 0, 8, 9, 11, 13, 67, 89]  
Even Numbers List: [12, 0, 8]  
Odd Numbers List: [1, 9, 11, 13, 67, 89]
```